

Fluorescent labelling and detection involve binding fluorescent dyes to biomolecules so as to visualise them through fluorescence imaging, such as lymph nodes in a pig. Theranostics helps in detecting target biomolecules like tumour cells for early diagnosis of cancer, tracking progress of tumour elimination, and localised treatment to specific sites.

Last year, researchers headed by a team at the Massachusetts Institute of Technology (MIT) created a microneedle platform using QDs, which can deliver vaccines as well as invisibly encode vaccination history directly in the skin for recordkeeping at the same time. The NIR emission detection was made possible by a specially equipped smartphone.

They can also work as optical indicators for the development of biosensors. For instance, Project BioSensing, a collaboration among Germany's Fraunhofer institutes (ISC and IME) and Leiden University's Institute of

Physics in the Netherlands, is aimed at developing advanced biosensors using quantum technology.

QDs also find other application areas such as solar cells, lasers, photovoltaic devices, computing, and so on. One such example is a new type of solar panel created by scientists at the National Research Nuclear University MEPhI, Russia based on hybrid material consisting of QDs and photosensitive protein. According to a report from Regal Intelligence, the global QDs market is expected to grow to US\$ 10,423.13 million by the end of 2025 at a CAGR of 22.06 per cent.

Although there are numerous chemical and physical methods available to synthesise QDs, biosynthetic methods enhance the biocompatibility and biostability of their structure through the utilisation of biomolecules such as proteins, enzymes, and nucleic acids for formation. Other ways include modifying the surface of available QDs and using artificial

cellular structures. In these types of processes, no toxic products are generated and the QDs obtained are suitable for biological and biomedical applications. This is important as toxic elements like cadmium, mercury, lead, and more can leak causing cell death and inflammation.

Several challenges in this realm still remain to be solved including environmental impact, toxicity, manufacturing costs, among others. This restricts their use to animal research. The issue is that biosynthesised QDs have a lower quantum yield and lesser uniformity in their size as compared to those synthesised through other methods. More research needs to be done in relation to the mechanisms and protocols for the scalability of high-quality, non-toxic QD production. In addition, the biosensors must have higher sensitivity to improve the reliability and efficiency of disease diagnosis in real-time.

—Ayushee Sharma

4 HOW 3D PRINTING IS TRANSFORMING BUSINESSES

3D printing has become an integral part of several sectors such as healthcare, aerospace, architecture, automotive and industrial goods. It has led to increase in the productivity and efficiency of the different sectors, and surely has a bright future for industries worldwide

Unlike traditional manufacturing, 3D printers can be used to make models multiple times by simply changing the algorithm, leading to faster production and decreased turnaround times. The three-dimensional digital model is converted to a physical object by adding multiple successive thin layers of materials like polymers, metals, ceramics, and the like. The quality of products is much better along with several customisation options. Although the cost of buying a 3D printer can be high based on the requirement, the reduced cost of constructing prototypes provides a quick return on investment.

Due to these advantages, students like those in engineer-

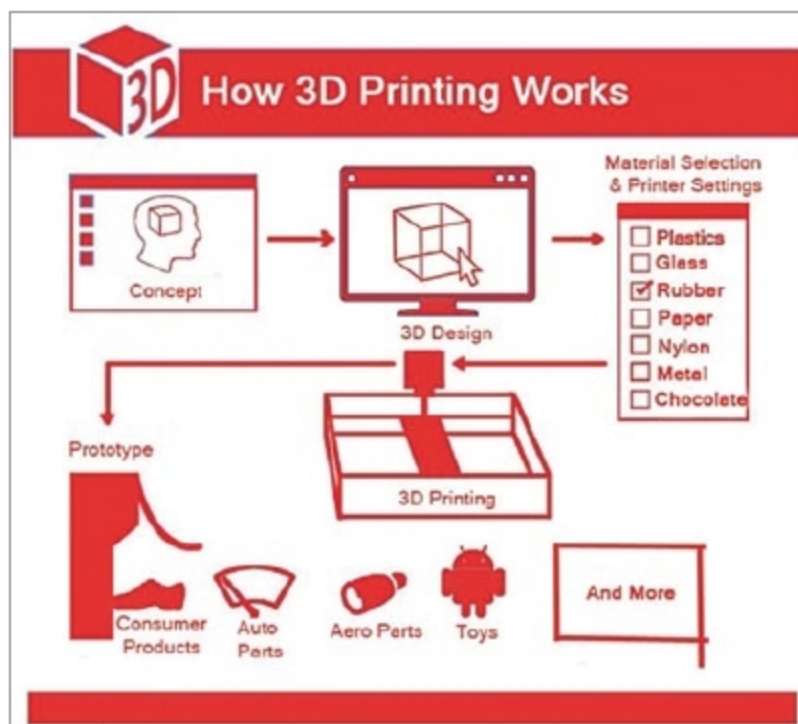
ing and medical, can test their designs and improve, while professionals like dentists can create durable moulds

that are resistant to pressure and temperature changes. Hence, it gives individuals the chance to experiment with

new materials and bring their vision to life, prior to making huge investments in production.

3D printing has become an integral part of different sectors such as healthcare, aerospace, architecture, automotive and industrial goods, among others, to increase productivity and efficiency in manufacturing. According to a report from research firm Wohler, the 3D printing market is estimated to grow to 35.6 billion dollars by 2024.

In one such instance, four companies—Air New Zealand, Microsoft, Moog and ST Engineering—teamed up to digitise the aerospace supply chain by



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